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(54) FIBER MEMBER AND HEAT EXCHANGER INCLUDING FIBER MEMBER

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(57) **ABSTRACT**

A fiber member is made of mutually entwined fibers, a surface of each of the fibers is coated with a material having a thermal conductivity higher than the fibers, and a porosity Pr (Pr=Vp/Vm×100) is 60% or more, where an apparent volume of the fiber member is Vm, and a void volume of the fiber member is Vp.

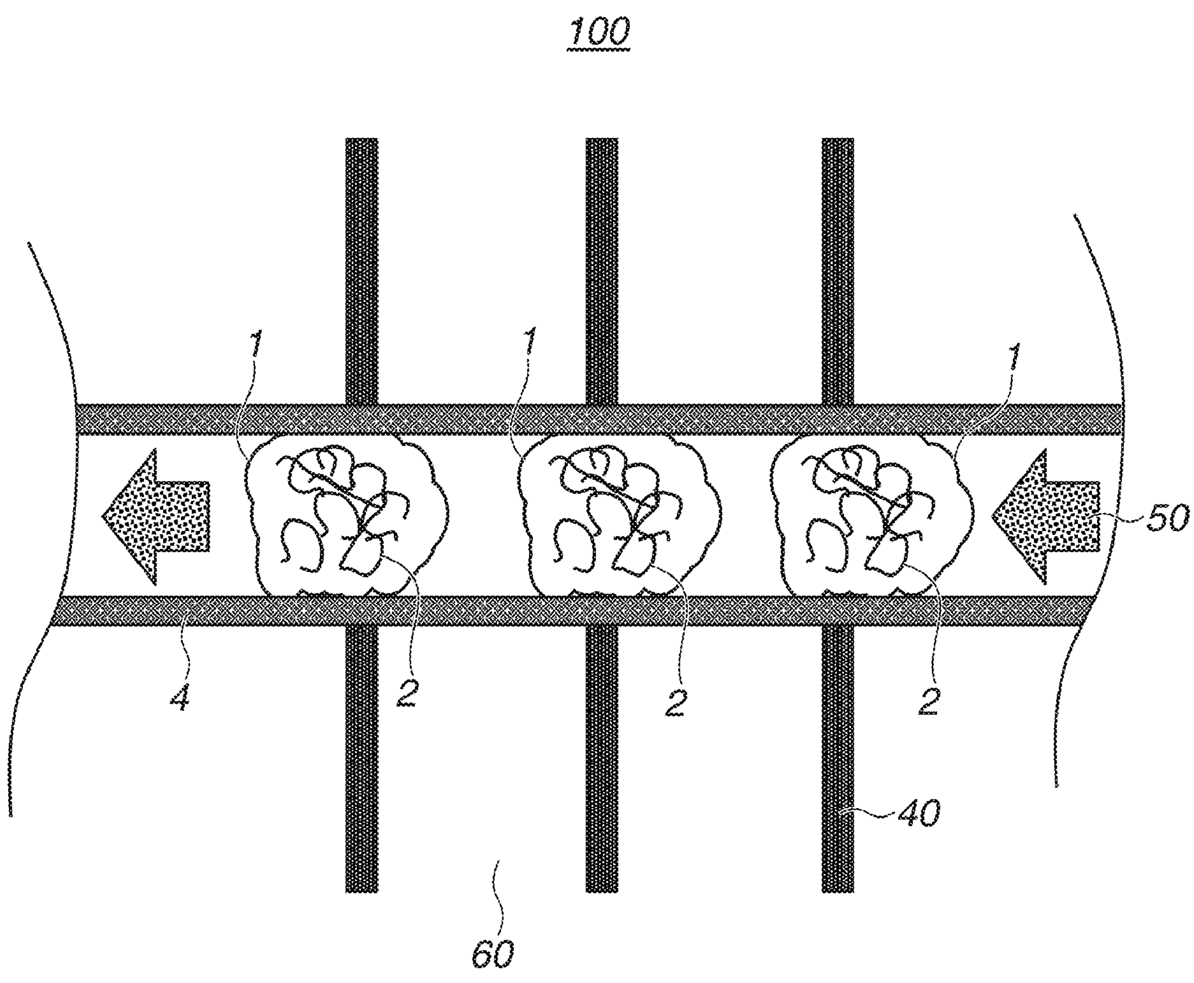


FIG.1

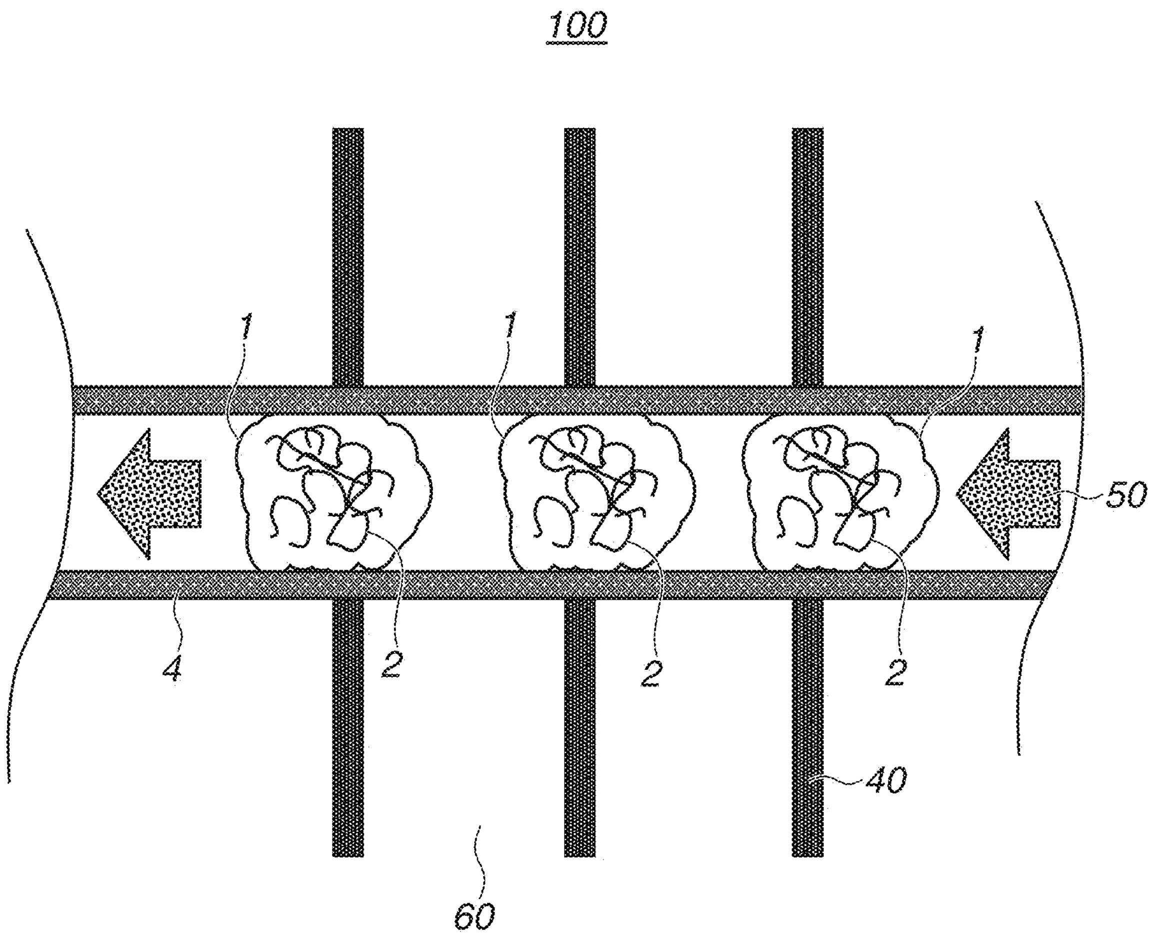


FIG.2

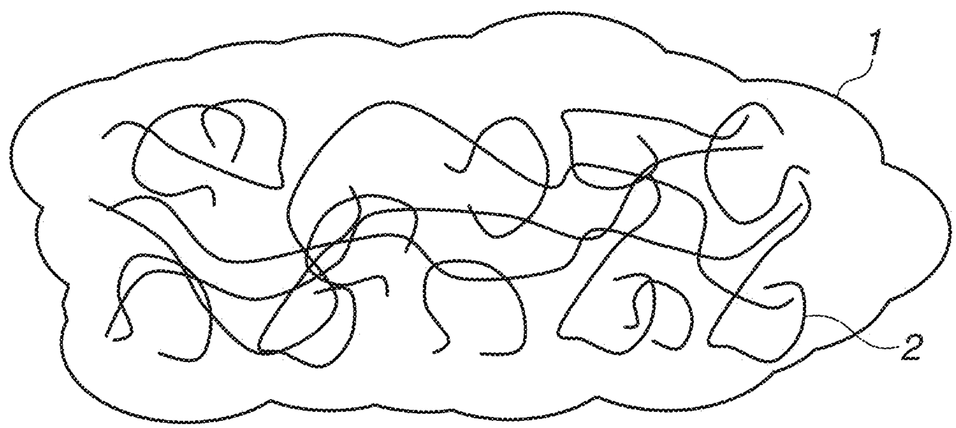


FIG.3A

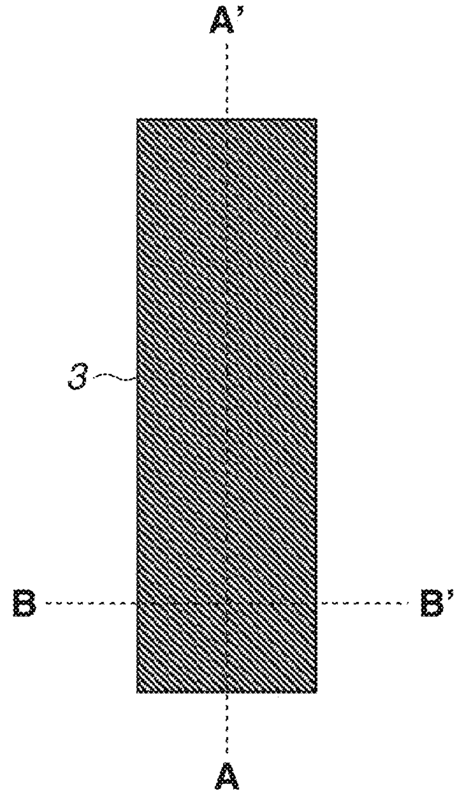


FIG.3B

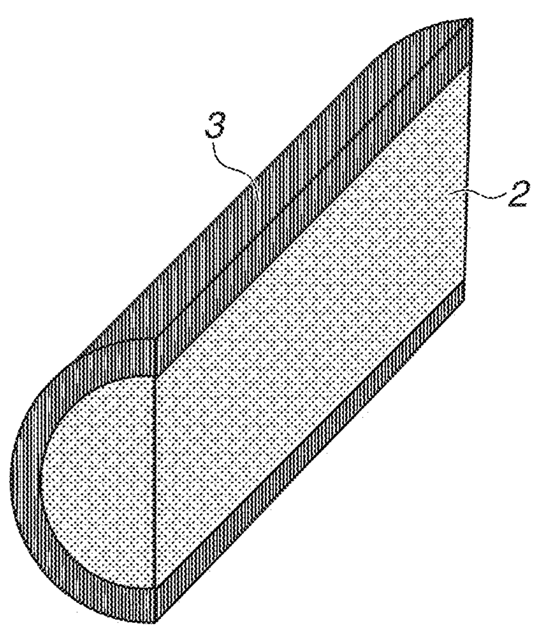


FIG.4

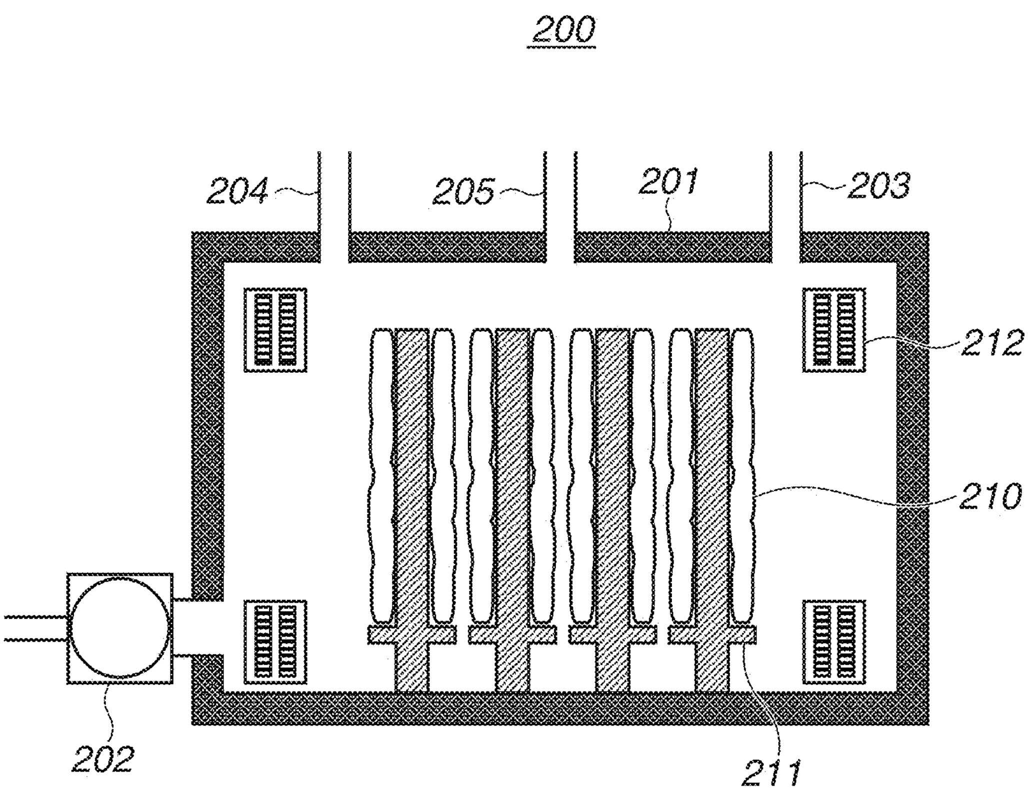
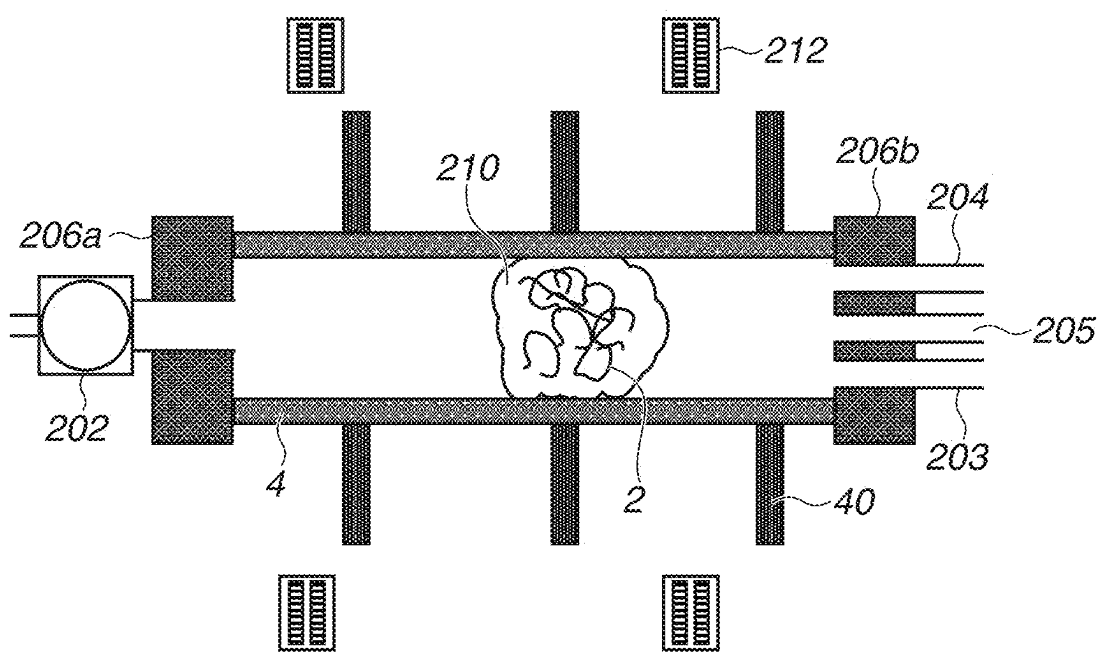


FIG.5



FIBER MEMBER AND HEAT EXCHANGER INCLUDING FIBER MEMBER

BACKGROUND

Field of the Disclosure

[0001] The present disclosure relates to a fiber member and a heat exchanger including the fiber member.

Description of the Related Art

[0002] A heat exchanger that performs a heat exchange using a difference in temperature between two fluids is widespread and used as a product such as an air conditioner and a water heater. The heat exchanger is known as an apparatus capable of efficiently heating and cooling targets due to its basic principle. As a main configuration of the heat exchanger, a method of flowing a refrigerant through a heat-transfer tube to exchange heat with a fluid outside the heat-transfer tube is employed.

[0003] Japanese Patent Application Laid-open No. 2009-270755 discusses a technique of forming a groove continuously extending in a spiral shape in an outer surface of the heat-transfer tube and forming a protrusion continuously extending in a spiral shape on an inner surface of the heat-transfer tube, to improve a heat exchanging performance in such a heat exchanger.

SUMMARY

[0004] According to an aspect of the present disclosure, a fiber member is made of mutually entwined fibers, wherein a surface of each of the fibers is coated with a material having a thermal conductivity higher than the fibers, and a porosity Pr ($Pr=Vp/Vm\times100$) is 60% or more, where an apparent volume of the fiber member is Vm, and a void volume of the fiber member is Vp.

[0005] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a schematic diagram illustrating a heat-transfer tube unit of a heat exchanger.

[0007] FIG. **2** is a schematic diagram illustrating a fiber member to be provided inside the heat-transfer tube unit.

[0008] FIG. **3**A and **3**B are an enlarged view and a cross-section view of a fiber of the fiber member, respectively.

[0009] FIG. **4** is a diagram illustrating an example of an atomic layer deposition (ALD) apparatus for coating the fiber with a material.

[0010] FIG. **5** is a diagram illustrating a modification example of a method of coating the fiber with the material.

DETAILED DESCRIPTION

[0011] From a perspective of carbon neutrality as a countermeasure against global warming in these years, and because of a rapid increase of resource prices, because a higher heat-exchange efficiency of a heat exchanger is demanded, the technique discussed in Japanese Patent Application Laid-open No. 2009-270755 is not enough for the demand, and further improvement of the heat-exchange efficiency is demanded.

[0012] The present disclosure is directed to a fiber member capable of improving a heat-exchange efficiency.

[0013] With reference to the attached drawings, exemplary embodiments of the present disclosure will be described in detail. However, the exemplary embodiments described below are merely examples of the present disclosure, and the present disclosure is not limited to the exemplary embodiments. In the descriptions below and drawings, common elements are assigned common reference symbols.

[0014] FIG. **1** is a schematic diagram illustrating a heat-transfer tube unit **100** of a heat exchanger according to an exemplary embodiment. A heat-transfer tube **4** is a tube made of copper, aluminum, or the like for flowing a refrigerant **50** therethrough, and heat exchange fins **40** are welded to its outer surface. As the refrigerant **50**, hydrofluorocarbon (HFC) or water can be used. A heat exchange is performed as heat is transmitted from the refrigerant **50** to an external fluid **60** via the heat-transfer tube **4** and the heat exchange fins **40**, and the heat-transfer tube unit **100** functions as a heat exchanger. In FIG. **1**, a mechanism for performing heating and cooling by compressing and expanding the refrigerant **50**, and a pump for sending the refrigerant **50** are not illustrated.

[0015] In the present exemplary embodiment, the heat-exchange efficiency between the heat-transfer tube **4** and the refrigerant **50** is improved, by inserting fiber members **1** (described below) inside the heat-transfer tube **4** so as to contact an inner wall of the heat-transfer tube **4**. In the exemplary embodiment illustrated in FIG. **1**, an example in which the three fiber members **1** are arranged inside the heat-transfer tube **4** is illustrated, but the number of the fiber members **1** and the positions thereof are not limited to those in the example, and they can be arranged as appropriate depending on a desired degree of the thermal efficiency.

[0016] By arranging the fiber members **1** so as to contact the inner wall of the heat-transfer tube **4**, the refrigerant **50** and the fiber members **1** in the heat-transfer tube **4** can contact each other in a wide surface area, and accordingly, the heat-exchange efficiency increases.

[0017] Each of the fiber members **1** is configured to include fibers **2**, and a pressure loss due to the fiber members **1** can be reduced by providing a porosity of a predetermined value or more. Further, it is possible to improve the heat-exchange efficiency without limiting the material of the fibers **2**, by coating the fibers **2** of the fiber members **1** with a material **3** having a thermal conductivity higher than the fibers **2**.

[0018] FIG. **2** is a schematic diagram illustrating one of the fiber members **1** provided inside the heat-transfer tube **4**. The fiber member **1** is made of a large number of the fibers **2** randomly entwined with each other. In order to improve the thermal efficiency, it is important for each of the fiber members **1** arranged inside the heat-transfer tube **4** to have a sufficient surface area for the heat exchange, and to be able to sufficiently reduce the pressure loss when the refrigerant **50** flows. For this reason, a porosity Pr of the fiber member **1** can be 60% or more, and further the porosity Pr can be 80% or more, from a viewpoint of the pressure loss.

[0019] The definition of the porosity Pr can be expressed by a ratio between an apparent volume Vm obtained from the outline of the fiber member **1**, i.e., the volume with the outline of the fiber member **1** being an external form, and a void volume Vp of the fiber member **1**, and expressed by a following formula (1).

$$\text{Porosity } Pr = Vp/Vm \times 100 \quad (1)$$

[0020] In a case where the fiber member **1** is inserted in the heat-transfer tube **4** and used, the porosity Pr can be 60% or more in a state where the fiber member **1** is arranged in the heat-transfer tube **4**. The surface area of the fiber member **1** for performing the heat exchange can be increased by making the fiber member **1** thick in a direction in which the refrigerant **50** flows through inside the heat-transfer tube **4**. The thickness can be 1 mm or more, or can be 2 mm or more, but the thickness can be less than 50 mm for easiness of arrangement.

[0021] FIG. **3**A is an enlarged view of one of the fibers **2** of the fiber member **1**, and FIG. **3**B is a cross-section diagram of the fiber **2** illustrated in FIG. **3**A, and is a diagram obtained by cutting the fiber **2** along a lengthwise direction A-A' cross-section and a radial direction B-B' cross-section illustrated in FIG. **3**A. Although it is not illustrated in FIG. **2**, the surface of the fiber **2** is entirely coated with the material **3**. In FIG. **3**, the shape of the fibers **2** is expressed in a linear manner, but actually is a shape with combined curvatures.

[0022] As the material of the fiber **2**, a carbon fiber, a glass fiber, and a resin fiber such as a polyester fiber, and/or a natural fiber such as a cellulose nanofiber can be selected and used as appropriate. The smaller the fiber diameter of the fiber **2** is, the larger the specific surface area of the fiber member **1** can be. Further, the smaller the fiber diameter is, the more advantageous it is for the pressure loss. Thus, the fiber diameter of the fiber **2** can be 1,000 nm or less, or can be 200 nm or less. Although there is a possibility that fibers with the fiber diameter of 1,000 nm or more are mixed because of the production difficulty, but it is possible to achieve a desired effect if a large amount of such fibers do not mix to a degree that the average diameter of the fibers **2** included in the fiber member **1** becomes more than 1,000 nm. More specifically, the fiber member **1** can be made of fibers with the average diameter of 1,000 nm or less, and further the average diameter can be 200 nm or less. On the other hand, since the strength of the fiber member **1** can be increased by intentionally mixing a small amount of fibers with a thick fiber diameter, the fibers can therefore be mixed, with the thick fiber diameter within the range of the average diameter described above, in the fibers **2** of the fiber member **1**.

[0023] A material with a thermal conductivity higher than that of the fiber **2** can be selected as the material **3** for coating the fiber **2**. As examples of the material **3**, metal, oxide, or both of them may be used, and as more specific examples, elementary Al, Cu, Ag, Au, or HfO_2, or their complex compositions can be used.

[0024] The film thickness of the material **3** can be about 0.5 nm or more and 100 nm or less so that the voids of the fiber member **1** do not reduce due to the coating. The fibers **2** located at the center portion of the fiber member **1** can be uniformly coated similar to the fibers **2** located at a peripheral portion of the fiber member **1**. In this case, the center portion of the fiber member **1** means an area S with a provisional center of gravity G as a center when a specific gravity inside the fiber member **1** is assumed to be constant, which has a figure similar to the outline of the fiber member **1** and 10% of the apparent volume Vm of the fiber member **1**. As the composition of the material **3**, the content rate of carbon (C) atoms can be 1.0 at % or more and 25 at % or less near the surface. This is because "C" derives from the "CH" group included in the material **3**, and since the hydrophilic property becomes insufficient in a case where the content rate of "C" is larger than 25 at %, there is a possibility that a good heat exchange cannot be achieved. The content rate of "C" can be analyzed using an X-ray photoelectron spectroscopy (XPS) apparatus.

Production Method of Fiber Member

[0025] The fiber member **1** may be produced by coating a single fiber of the fibers **2** with the material **3** and then entwining the fibers **2** together, or by entwining the fibers **2** together before being coated together, and then coating the fibers **2** with the material **3**. The method of coating the fibers **2** with the material **3** is not specifically limited, and a coting method using a known technique, such as wet coating, sputtering, vapor-deposition, ion plating, chemical vapor deposition (CVD), and atomic layer deposition (ALD) methods can be selected. Examples of the CVD method include a plasma CVD method using plasma, and a heat CVD method using heat.

[0026] From a viewpoint of a stable production, the fibers **2** can be coated with the material **3** by using an ALD method after entwining the fibers **2** together and before being coated with the material **3**.

[0027] The ALD method is a method capable of forming a thin film in atomic layer unit while alternately repeating intake and evacuation of two or more kinds of material gases and reacting the material adsorbed into a film formation target surface, using, for example, a method discussed in Japanese Unexamined Patent Application Publication No. 2009-525406.

[0028] FIG. **4** illustrates an example of an ALD apparatus **200** performing the ALD method for coating the fibers **2** with the material **3**. A vacuum pump **202** for vacuum exhaust, a precursor supply line **203**, a reactive gas supply line **204**, and a purge gas supply line **205** are connected to a metallic vacuum chamber **201** of the ALD apparatus **200**. The lines connected to the vacuum chamber **201** are each provided with a valve for flow control, and the like, but they are not illustrated in FIG. **4**.

[0029] In the vacuum chamber **201**, holders **211** for holding the fiber members **210** each made by entwining the fibers **2** together that are going to be coated with the material **3**, and heaters **212** for heating the fiber members **210** are included. The fiber members **210** before being coated are set to the respective holders **211**, and the vacuum chamber **201** is vacuumed to about 1 Pa. The heaters **212** are heated to heat the fiber members **210** to an appropriate temperature. The heating temperature can be changed to match the used precursor or the reactive gas, and is about 100 to 500° C. The reactive gas is taken in when the temperature of the fiber members **210** reaches a target temperature, and the gas intake is stopped after taking in the reactive gas for a predetermined time period. Then, the purge gas is taken in for a predetermined time period, to eliminate the unnecessary precursor and the cracked gas. Then, the reactive gas is taken in for a predetermined time period to coat the fibers **2** with a desired composition of the material **3**. Then, the purge gas is taken in to eliminate the unnecessary gas. By repeating the series of cycles to coat the fibers **2** with the material **3** to a desired film thickness, the fiber member **1** is produced.

[0030] In a case where "Cu" is coated on the fibers **2** as the material **3**, Cu (4-Dimethylaminopyridine (DMAP)) **2** and hydrazine can be used as the precursor and the reactive gas, respectively. Depending on the coating condition, the material **3** includes "C" of 1 to 20 at %. With "C", the adhesiveness of the material **3** to the fibers **2** increases. The detailed reason is not known, but it is thought that the coating density lowers by including "C", which reduces the internal stress.

[0031] After the coating, the heating of the heater **212** is stopped, and the vacuum chamber **201** is air-released after the vacuum chamber **201**, the holders **211**, and the coated fibers **2** are sufficiently cooled, to take out the fiber member **1**.

[0032] By setting the fiber member **1** produced in this way in the heat-transfer tube **4** of the heat exchanger so as to contact the inner wall of the heat-transfer tube **4**, the heat-exchange efficiency can be improved.

Modification Example of Fiber Member Production Method

[0033] In the method illustrated in FIG. **4**, the fiber members **1** are arranged inside the ALD apparatus to be coated with the material **3**, but the fibers **2** may be coated with the material **3** using the ADL method in a state where the fiber members **1** are arranged inside the heat-transfer tube **4**.

[0034] FIG. **5** illustrates a state where the coating with the material **3** is performed with the heat-transfer tube **4** being assumed as a vacuum chamber of the ALD apparatus in a state where the fiber member **210** made of the fibers **2** entwined with each other to be coated with the material **3** is set in the heat-transfer tube **4**. In this case, open ends of the heat-transfer tube **4** are connected with vacuum flanges **206***a* and **206***b*. The vacuum flange **206***a* connected with the vacuum pump **202** for vacuum exhaust, the precursor supply line **203**, the reactive gas supply line **204**, and the vacuum flange **206***b* connected with the purge gas supply line **205** are connected to the heat-transfer tube **4**. The heat-transfer tube **4** is vacuumed by the vacuum pump **202**, and the heat-transfer tube **4** and the fiber member **210** are heated by the heater **212** placed outside the heat-transfer tube **4**. Then, the coating with the material **3** is performed in a similar manner to that by the ALD apparatus in FIG. **4**.

[0035] With the modification example illustrated in FIG. **5**, since the fiber member **210** and the heat-transfer tube **4** are coated with the material **3** at a same time, the contact portion between the fiber member **210** and the heat-transfer tube **4** is coated and covered by the material **3**. By covering the contact portion with the material **3** in this way, the heat-exchange efficiency can further be improved.

Evaluation of Examples and Comparative Examples

[0036] The present disclosure will be described specifically using examples and comparative examples. However, the range of the present disclosure is not limited to such examples.

[0037] As the heat-transfer tube **4**, a copper cylindrical tube with an outer diameter of 12 mm, an inner diameter of 10 mm, and a length of 1 m was used. As the refrigerant **50**, a 15° C. refrigerant **50** R32 (HFC-32) was circulated at a flow rate of 0.01 kg/min, and an amount of heat-exchanged heat quantity was evaluated. The room temperature and the heat-transfer tube temperature of the heat-transfer tube **4** at an examination start time were set to 25° C.

[0038] A first example will be described. The fibers **2** with an average fiber diameter of 900 nm were coated with "Cu" of 100 nm as the material **3** using the ALD method. When the composition was analyzed using the XPS, "C" of 5 at % was included near the coating surface. The fiber member **1** was set inside the heat-transfer tube **4** so that the porosity of the fiber member **1** became 60%, and the thickness of the fiber member **1** in a direction in which the refrigerant **50** inside the heat-transfer tube **4** flew became 5 mm.

[0039] Next, a second example will be described. The fibers **2** with an average fiber diameter of 200 nm were coated with "Cu" of 100 nm as the material **3** using the ALD method. The fiber member **1** was set inside the heat-transfer tube **4** so that the porosity of the fiber member **1** became 80%, and the thickness of the fiber member **1** in a direction in which the refrigerant **50** inside the heat-transfer tube **4** flew became 5 mm.

[0040] Next, a comparative example 1 will be described. The heat-transfer tube **4** without the fiber member **1** arranged was prepared.

[0041] Next, a comparative example 2 will be described. Using the fiber member **210** made by entwining the fibers **2** together with an average fiber diameter of 900 nm (fiber member before coated with the material **3**), and the fiber member **210** was arranged inside the heat-transfer tube **4** so that the porosity of the fiber member **210** became 60%, and the thickness of the fiber member **210** in a direction in which the refrigerant **50** inside the heat-transfer tube **4** flew became 5 mm.

[0042] Parameters and heat-exchange efficiencies are illustrated in Table 1 in a collective manner. It was confirmed that the heat-exchange efficiencies in the first and second examples were improved compared with the comparative example 1. On the other hand, no difference was found between the heat-exchange efficiencies of the comparative examples 1 and 2. More specifically, from the result of this confirmation, it was confirmed that the heat-exchange efficiency could be improved by arranging, inside the heat-transfer tube **4**, the fiber member **1** with its surface coated by the material **3** of a substance having a thermal conductivity higher than the fibers **2**. It was also confirmed that the heat-exchange efficiency could be further improved by increasing the porosity of the fiber member **1**. Thus, since the heat-exchange efficiency can be improved by setting the fiber member according to the present disclosure to the heat-transfer tube of the heat exchanger, the reduction of the consumed energy of the heat exchange apparatus such as an air conditioning apparatus can be expected.

TABLE 1

		Example 1	Example 2	Comparative example 1	Comparative example 2
Presence (YES)/Absence (NO) of Fiber Member 1		YES	YES	NO	YES
Fiber Member 1 Parameter	Coating	Cu	Cu	—	Uncoated
	Porosity (%)	60	80	—	60
	Average fiber diameter (nm)	900	200	—	900
	Thickness (mm)	5	5	—	5

TABLE 1-continued

	Example 1	Example 2	Comparative example 1	Comparative example 2
Heat-exchange Efficiency Evaluation Ranking	2	1	3	3

[0043] In this way, it is possible to provide a fiber member capable of improving a heat-exchange efficiency.

[0044] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0045] This application claims the benefit of Japanese Patent Application No. 2024-030011, filed Feb. 29, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A fiber member made of mutually entwined fibers,

wherein a surface of each of the fibers is coated with a material having a thermal conductivity higher than the fibers, and

wherein a porosity Pr (Pr=Vp/Vm×100) is 60% or more, where an apparent volume of the fiber member is Vm, and a void volume of the fiber member is Vp.

2. The fiber member according to claim **1**, wherein the material coating the surface of each of the fibers comprises at least one of Al, Cu, Ag, Au, and HfO_2.

3. The fiber member according to claim **1**, wherein the material coating the surface of each of the fibers comprises carbon atoms of 1 at % or more and 25 at % or less.

4. The fiber member according to claim **1**, wherein an average diameter of the fibers of the fiber member is 1,000 nm or less.

5. The fiber member according to claim **1**, wherein the fibers comprise a carbon fiber, a glass fiber, a resin fiber and/or a natural fiber.

6. A heat-transfer tube unit comprising:

a heat-transfer tube; and

the fiber member according to claim **1** inside the heat-transfer tube,

wherein a heat exchange is performed by flowing a refrigerant in the heat-transfer tube.

7. The heat-transfer tube unit according to claim **6**, wherein a thickness of the fiber member is 1 mm or more in a direction in which the refrigerant in the heat-transfer tube flows.

8. The heat-transfer tube unit according to claim **6**, wherein an inner wall of the heat-transfer tube is in contact with the fiber member.

9. The heat-transfer tube unit according to claim **6**, wherein an inner wall of the heat-transfer tube is coated with a same material as the material coating the surface of each of the fibers.

10. The heat-transfer tube unit according to claim **6**, comprising heat exchange fins welded to an outer surface of the heat-transfer tube.

11. The heat-transfer tube unit according to claim **6**, wherein the heat-transfer tube is made of copper or aluminum.

12. A heat exchanger comprising:

the heat-transfer tube unit according to claim **6**; and

a refrigerant in the heat-transfer tube.

13. A production method of producing a fiber member to be arranged inside a heat-transfer tube for performing a heat exchange by flowing a refrigerant through the heat-transfer tube, the production method comprising:

preparing fibers;

coating surfaces of fibers with a material having a thermal conductivity higher than the fibers; and

entwining the fibers together.

14. The production method according to claim **13**, wherein the coating is performed after the entwining.

15. The production method according to claim **13**, wherein the coating is performed before the entwining.

16. The production method according to claim **13**, wherein the coating is performed using an atomic layer deposition method.

17. The production method according to claim **13**, wherein the coating is performed inside the heat-transfer tube.

18. The production method according to claim **13**, wherein the coating includes coating an inner wall of the heat-transfer tube with a material same as the material coating the surfaces of fibers.

19. The production method according to claim **13**, wherein the fibers comprise a carbon fiber, a glass fiber, a resin fiber and/or a natural fiber.

20. The production method according to claim **13**, wherein the material coating the surface of each of the fibers comprises at least one of Al, Cu, Ag, Au, and HfO_2.

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